

# D-SCRAMBLED CANTOR SETS

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ABSTRACT. For all  $d \geq 3$ , we provide an example of a homeomorphism of Cantor space that has an uncountable  $d$ -scrambled set but no  $d$ -scrambled Cantor set.

## INTRODUCTION

Suppose that  $X$  is a metric space and  $f: X \rightarrow X$ . We say that  $x \in X^d$  is *proximal* if  $\liminf_{n \rightarrow \infty} \max_{i < j < d} d_X(f^n(x(i)), f^n(x(j))) = 0$ , *separated* if  $\limsup_{n \rightarrow \infty} \min_{i < j < d} d_X(f^n(x(i)), f^n(x(j))) > 0$ , and *Li–Yorke* if it is proximal and separated. A set  $Y \subseteq X$  is  *$d$ -scrambled* if every injective sequence in  $Y^d$  is Li–Yorke, and  $f$  is *Li–Yorke  $d$ -chaotic* if there is an uncountable  $d$ -scrambled set  $Y \subseteq X$ .

In [GGM21], it was shown that every Li–Yorke 2-chaotic continuous function on a Polish space has a 2-scrambled Cantor set. Here we show:

**Theorem 1 (ZFC).** *Suppose that  $d \geq 3$ . Then there is a Li–Yorke  $d$ -chaotic homeomorphism of  $2^{\mathbb{N}}$  with no  $d$ -scrambled Cantor set.*

A *graph* on  $X$  is an irreflexive symmetric set  $G \subseteq X \times X$ . Let  $\text{Hyp}^d(G)$  be the  $d$ -ary relation on  $X$  of all injective sequences  $x \in X^d$  such that  $G \cap (x[d] \times x[d]) \neq \emptyset$ . An *embedding* of a  $d$ -ary relation  $R$  on  $X$  into a  $d$ -ary relation  $S$  on  $Y$  is an injection  $\pi: X \rightarrow Y$  such that  $x \in R \iff \pi \circ x \in S$  for all  $x \in X^d$ . The *saturation* of a set  $Y \subseteq X$  under a bijection  $T: X \rightarrow X$  is given by  $[Y]_T = \bigcup_{n \in \mathbb{Z}} T^n[Y]$ . A subset of a topological space is  $\Pi_2^0$  if it is a countable intersection of open sets, and  $\Sigma_3^0$  if it is a countable union of  $\Pi_2^0$  sets. The primary new technical result underlying our proof of Theorem 1 is the following:

**Theorem 2 (ZF + DC).** *Suppose that  $d \geq 3$  and  $G$  is a  $\Sigma_3^0$  graph on  $2^{\mathbb{N}}$ . Then there is a continuous embedding  $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  of  $\text{Hyp}^d(G)$  into the  $d$ -ary Li–Yorkeness relation of a homeomorphism  $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  for which there is a unique point of  $2^{\mathbb{N}}$  not in the  $T$ -saturation of  $\pi[2^{\mathbb{N}}]$ .*

In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

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## 1. EMBEDDING HYPERGRAPHS

Define  $S: 2^{\mathbb{Z}} \rightarrow 2^{\mathbb{Z}}$  by  $S(c)(i) = c(i+1)$  for all  $i \in \mathbb{Z}$  and  $c \in 2^{\mathbb{Z}}$ . Theorem 2 follows from [Kec95, Proposition 7.4] and:

**Theorem 1.1** (ZF + DC). *Suppose that  $d \geq 3$  and  $G$  is a  $\Sigma_3^0$  graph on  $2^{\mathbb{N}}$ . Then there exist an  $S$ -invariant perfect set  $F \subseteq 2^{\mathbb{Z}}$ , whose finite sequences are proximal, and a continuous embedding  $\pi: 2^{\mathbb{N}} \rightarrow F$  of  $\text{Hyp}^d(G)$  into the  $d$ -ary separation relation for which  $0^{\mathbb{Z}}$  is the unique point of  $F$  not in the  $S$ -saturation of  $\pi[2^{\mathbb{N}}]$ .*

*Proof.* For all  $s \in 2^{<\mathbb{N}}$ , define  $\mathcal{N}_s = \{c \in 2^{\mathbb{N}} \mid \forall i < |s| \ s(i) = c(i)\}$ . For all  $S \subseteq 2^{<\mathbb{N}}$ , define  $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$ . For all  $\ell \in \mathbb{N}$  and  $p: 2^\ell \rightarrow d$ , define  $\ell_p = \ell$  and  $\mathcal{N}_p = \prod_{i < d} \mathcal{N}_{p^{-1}(\{i\})}$ . We say that  $p$  is a *near bijection* if it is surjective and there is a unique  $k_p < d$  for which  $|p^{-1}(\{k_p\})| > 1$ . For all  $k \in d \setminus \{k_p\}$ , let  $p^{-1}(k)$  denote the unique element of  $p^{-1}(\{k\})$ .

Fix  $\Pi_2^0$  graphs  $G_n$  on  $2^{\mathbb{N}}$  for which  $G = \bigcup_{n \in \mathbb{N}} G_n$ , and decreasing sequences  $(G_{m,n})_{m \in \mathbb{N}}$  of open graphs whose intersection is  $G_n$ . Set  $H_{m,n} = \text{Hyp}^d(G_{m,n})$  and  $H_n = \bigcap_{m \in \mathbb{N}} H_{m,n}$ . Then  $\text{Hyp}^d(G) = \bigcup_{n \in \mathbb{N}} H_n$ . For all  $m, n \in \mathbb{N}$ , let  $P_{m,n}$  denote the set of all near bijections  $p$  for which  $\ell_p \geq m$ ,  $p(0^{\ell_p}) = 0$ , and  $\mathcal{N}_p \subseteq H_{m,n}$ . Let  $Q_{m,n}$  denote the set of all  $q \in P_{m,n}$  for which there is no  $p \in P_{m,n}$  such that  $k_p = k_q$ ,  $\ell_p < \ell_q$ , and  $p^{-1}(k) = q^{-1}(k) \upharpoonright \ell_p$  for all  $k \in d \setminus \{k_p\}$ .

Fix an infinite set  $I \subseteq \mathbb{N} \setminus \{0\}$  such that  $|I \cap (I - n)| < \aleph_0$  for all  $n > 0$ , and a bijection  $\phi: I \rightarrow \mathbb{N} \cup \bigcup_{m,n \in \mathbb{N}} \{m\} \times \{n\} \times Q_{m,n}$  such that  $I \cap [i, 2i + (d-2)(\phi(i)(1) + 1)] = \{i\}$  for all  $i \in I \setminus \phi^{-1}(\mathbb{N})$ . Define a continuous function  $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{Z}}$  by setting  $\pi(c)(i) = 1$  if and only if

- (1)  $i = 0$ ;
- (2)  $i \in I$ ,  $\phi(i) \in \mathbb{N}$ , and  $c(\phi(i)) = 1$ ; or
- (3)  $\exists 0 < k < d \exists m, n \in \mathbb{N} \exists q \in Q_{m,n}$   
 $((m, n, q) = \phi(i - (k-1)(n+1)) \text{ and } q(c \upharpoonright \ell_q) = k)$ .

**Lemma 1.2.** *The function  $\pi$  is injective.*

*Proof.* Suppose that  $c, c' \in 2^{\mathbb{N}}$  are distinct. Then there exists  $n \in \mathbb{N}$  with  $c(n) \neq c'(n)$ . Fix  $i \in I$  for which  $\phi(i) = n$ . As  $\pi(c)(i) = c(n)$  and  $\pi(c')(i) = c'(n)$ , it follows that  $\pi(c)(i) \neq \pi(c')(i)$ , so  $\pi(c) \neq \pi(c')$ .  $\square$

The *product metric* on  $2^{\mathbb{Z}}$  is given by  $\rho(c, c') = \sum_{i \in \mathbb{Z}} |c(i) - c'(i)|/2^{|i|}$  for all  $c, c' \in 2^{\mathbb{Z}}$ . A *homomorphism* from a  $d$ -ary relation  $R$  on a set  $Y$  to a  $d$ -ary relation  $S$  on a set  $Z$  is a function  $\pi: Y \rightarrow Z$  such that  $\pi \circ y \in S$  for all  $y \in R$ . Define  $N_i = \{i\}$  for all  $i \in \phi^{-1}(\mathbb{N})$  and  $N_i = [i, i + (d-2)(\phi(i)(1) + 1)]$  for all  $i \in I \setminus \phi^{-1}(\mathbb{N})$ .

**Lemma 1.3.** *The function  $\pi$  is a homomorphism from  $\text{Hyp}^d(G)$  to the  $d$ -ary separation relation.*

*Proof.* Suppose that  $c \in \text{Hyp}^d(G)$ , and fix  $n \in \mathbb{N}$  and  $j, k < d$  such that  $c(j) G_n c(k)$ . By permuting the coordinates of  $c$ , we can assume that either  $c(0) = 0^{\mathbb{N}}$  or both  $0^{\mathbb{N}} \notin c[d]$  and  $j, k > 0$ . Fix any  $k' \in d \setminus \{j, k\}$  in the former case, and set  $k' = 0$  in the latter. If  $m \in \mathbb{N}$  is sufficiently large that  $\text{proj}_{2^m} \circ c$  is injective and  $0^{\mathbb{N}} \notin c[d] \implies 0^m \notin (\text{proj}_{2^m} \circ c)[d]$ , then there is a least  $\ell \geq m$  for which there is a near bijection  $q \in P_{m,n}$  extending  $(\text{proj}_{2^\ell} \circ c)^{-1}$  such that  $k_q = k'$  and  $\ell_q = \ell$ . As  $q \in Q_{m,n}$ , there exists  $i \in I$  with  $\phi(i) = (m, n, q)$ . Then  $\text{proj}_{2^{N_i}} \circ \pi \circ c$  is injective, so  $\min_{j < k < d} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{(d-2)(n+1)}$ .  $\square$

**Lemma 1.4.** *The function  $\pi$  is a homomorphism from  $(2^{\mathbb{N}})^d \setminus \text{Hyp}^d(G)$  to the complement of the  $d$ -ary separation relation.*

*Proof.* Suppose that  $c \in (2^{\mathbb{N}})^d$  and  $\pi \circ c$  is separated. Fix  $\epsilon > 0$  such that  $I' = \{i' \in \mathbb{N} \mid \min_{j < k < d} \rho(S^{i'}(\pi(c(j))), S^{i'}(\pi(c(k)))) \geq \epsilon\}$  is infinite, as well as  $r \in \mathbb{N}$  such that  $\sum_{|i| > r} 1/2^{|i|} < \epsilon$ . By throwing out finitely many elements of  $I'$ , we can assume that for all  $i' \in I'$ , there is at most one  $i \in I$  such that  $[i' - r, i' + r] \cap N_i \neq \emptyset$ . Our choice of  $r$  then yields a unique such  $i \in I$ , in which case  $\text{proj}_{2^{N_i}} \circ \pi \circ c$  is injective. As  $d \geq 3$ , there exist  $m, n \in \mathbb{N}$ ,  $q \in Q_{m,n}$ , and  $\sigma \in \text{Sym}(d)$  for which  $(d-2)(n+1) \leq 2r$ ,  $\phi(i) = (m, n, q)$ , and  $c \circ \sigma \in \mathcal{N}_q$ . By passing to an infinite subset of  $I'$ , we can assume that the values of  $k_q$ ,  $n$ , and  $\sigma$  are independent of  $i' \in I'$ . The minimality in the definition of  $Q_{m,n}$  from  $P_{m,n}$  then ensures that elements of  $I'$  corresponding to the same value of  $m$  also correspond to the same value of  $i$ . As the distance between any two such elements of  $I'$  is at most  $2r$ , it follows that  $c \in H_{m,n}$  for infinitely many  $m \in \mathbb{N}$ , so  $c \in H_n$ , thus  $c \in \text{Hyp}^d(G)$ .  $\square$

The closure  $F$  of the  $S$ -saturation of  $\pi[2^{\mathbb{N}}]$  is clearly perfect.

**Lemma 1.5.** *The sequence  $0^{\mathbb{Z}}$  is the unique element of  $F$  not in  $[\pi[2^{\mathbb{N}}]]_S$ .*

*Proof.* Suppose that  $c \in F \setminus [\pi[2^{\mathbb{N}}]]_S$ . Then there exist a sequence  $(c_n)_{n \in \mathbb{N}}$  of elements of  $2^{\mathbb{N}}$ ,  $\epsilon \in \{\pm 1\}$ , and a strictly increasing sequence  $(i_n)_{n \in \mathbb{N}}$  of natural numbers such that  $S^{\epsilon i_n}(\pi(c_n)) \rightarrow c$ . If  $\epsilon = -1$ , then clearly  $c = 0^{\mathbb{Z}}$ . If  $\epsilon = 1$  and  $c \neq 0^{\mathbb{Z}}$ , then our choice of  $I$  ensures that  $c$  is in the  $S$ -orbit of  $\pi(0^{\mathbb{N}})$ , a contradiction.  $\square$

Lemma 1.5 implies that the union of the supports of finitely many elements of  $F$  is contained in a finite union of translates of  $\{0\} \cup \bigcup_{i \in I} N_i$ . As the defining property of  $I$  ensures that such sets have arbitrarily long gaps, every finite sequence of elements of  $F$  is proximal.  $\square$

A set  $Y \subseteq X$  is a  $G$ -clique if all distinct elements of  $Y$  are  $G$ -related, and  $G$ -independent if no elements of  $Y$  are  $G$ -related.

**Proposition 1.6** (ZF + DC). *Suppose that  $d \geq 2$ ,  $X$  and  $Y$  are Polish spaces,  $G$  is a Borel graph on  $X$ ,  $T: Y \rightarrow Y$  is a Borel automorphism,  $B \subseteq Y$  is an uncountable Borel  $d$ -scrambled set,  $\pi: X \rightarrow Y$  is an injective Borel homomorphism from  $X^d \setminus \text{Hyp}^d(G)$  to the complement of the  $d$ -ary Li–Yorkeness relation, and the saturation of  $\pi[X]$  is co-countable. Then  $G$  has a perfect clique.*

*Proof.* Fix  $i \in \mathbb{Z}$  for which  $B \cap (T^i \circ \pi)[X]$  is uncountable. The perfect set theorem and Galvin’s Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set  $P \subseteq (T^i \circ \pi)^{-1}(B)$  that is either a  $G$ -clique or  $G$ -independent. But  $G$ -independence would contradict the fact that  $B$  is  $d$ -scrambled.  $\square$

## 2. $D$ -SCRAMBLED SETS

A subset of a topological space is  $\Sigma_2^0$  if its complement is  $\Pi_2^0$ .

*Proof (of Theorem 1).* By [KV12, Theorem 2.1] and a simple Baire category argument, there is a  $\Sigma_2^0$  graph  $G$  on  $2^{\mathbb{N}}$  with an uncountable clique  $Y \subseteq 2^{\mathbb{N}}$  but no perfect clique. By Theorem 2, there is a continuous embedding  $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  of  $\text{Hyp}^d(G)$  into the  $d$ -ary Li–Yorkeness relation of a homeomorphism  $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  for which the saturation of  $\pi[2^{\mathbb{N}}]$  is co-countable. Then  $\pi[Y]$  is an uncountable  $d$ -scrambled set, but there is no perfect  $d$ -scrambled set by Proposition 1.6.  $\square$

Let  $\mathfrak{c}$  denote the cardinality of the continuum.

**Theorem 2.1** (ZFC +  $\neg\text{CH}$ ). *Suppose that  $d \geq 3$ . Then it is consistent that there is a homeomorphism  $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  with a  $d$ -scrambled set of cardinality  $\mathfrak{c}$  but no perfect  $d$ -scrambled set.*

*Proof.* By [She99, Theorem 1.13], we can assume that there is a  $\Sigma_2^0$  graph  $G$  on  $2^{\mathbb{N}}$  with a clique of cardinality  $\mathfrak{c}$  but no perfect clique. The proof of Theorem 1 therefore yields the desired result.  $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

**Theorem 2.2** (ZFC +  $\neg\text{CH}$ ). *Suppose that  $d \geq 3$ . Then it is consistent that every Borel function on a Polish space with a  $d$ -scrambled set of cardinality  $\aleph_2$  has a  $d$ -scrambled perfect set.*

*Proof.* By [KS03, Proposition 3.4], we can assume that every Borel  $d$ -ary hypergraph on a Polish space with a clique of cardinality  $\aleph_2$  has a perfect clique. But  $d$ -ary Li–Yorkeness is a Borel hypergraph.  $\square$

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